

Feasibility study of the digital tool Max for the patient-provided medication list in the medication reconciliation process prior to hospitalisation: patient willingness and usability, time saved and reliability

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ABSTRACT

Purpose More than 20% of prescription errors in hospitals are due to an incomplete medication history. Medication reconciliation is a solution to decrease unintentional discrepancies between medications taken at home and hospital prescriptions. It is a normalised clinical activity but it is time consuming. Medication reconciliation usually uses three sources of information for an optimised medical synthesis, one of which is the patient. A conversational robot for patients could be a solution to assist. Numerous digital applications are designed for patients and need to be tested for usability, satisfaction, reliability and time saved.

Method We analysed Max, a conversational robot for patients scheduled for surgery in Toulouse University Hospital, using routinely collected health data in three successive steps. We examined willingness, compliance and patient satisfaction of usability with a Likert questionnaire and measured the time spent with Max and without. Finally, the reliability has been explored.

Results The three successive observational steps were assessment of willingness and compliance (79 patients), time saved (61 patients) and reliability of the tool (68 patients). 71% agreed to use Max after a telephone call but only 73% of patients completed Max entirely. Max was well received and the overall satisfaction of usability was high for ease of use, readability, relevance and number of questions. Max saved a few minutes by optimised medical synthesis compared with a conventional telephone call. However, the reliability appeared to be lower than the human conventional telephone call. Randomised controlled trials are needed to confirm this feasibility study.

Conclusion Max was appreciated by patients and appeared to be suitable for assisting pharmacists in medication reconciliation. The tool established the list of treatments taken by the patient at home but reliability appeared to be lower than a conventional telephone call, recommending a 'double check' on the patient's arrival.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ The source of information from patients during medication reconciliation is paramount. Max is neither known nor deployed as a patient source; its acceptability among patients and its reliability among pharmacists need to be explored.

WHAT THIS STUDY ADDS

⇒ This study provides insights into the willingness of patients to use Max at home to provide their medication list, their satisfaction with its usability and, for pharmacists, the accuracy of the information gathered as well as a trend towards time saved.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This feasibility study paves the way for randomised controlled trials to support the routine use of Max.

medications taken at home through primary care and the first prescription in hospital (FPH).^{3,4} MR associated with other activities in clinical pharmacy can decrease the morbi-mortality for patients in many different clinical situations.⁵ Furthermore, MR can decrease unscheduled rehospitalisations at 30 days, as well as medical consultations⁶ and preventable adverse events at 30 days post-hospitalisation.⁷ However, MR is a highly time-consuming process, which usually takes more than 30 min per patient for the entire process.⁸ According to the literature, the mean time is 20–79 min.⁹ In fact, there are two main steps in MR. The first, optimised medical synthesis (OMS), is the summary of medications actually taken by a patient for which information is collected from several sources. The second is a comparison between the OMS and the FPH in order to detect unintentional discrepancies.

Electronic devices are commonly used in many areas of daily life such as e-Health.^{10–12} Conversational robots for patients have been introduced more frequently over the past decade. In fact, humanoid robots have recently been tested in MRI care services and have been well accepted by patients whose knowledge about MRI has thereby increased.¹³ The Food and Drug Administration

INTRODUCTION

Background

More than 20% of prescription errors in hospitals are due to an incomplete medication history.¹ Medication reconciliation (MR) is used in clinical pharmacy to decrease medication risk, particularly during transitions in care to hospital.² MR on admission to hospital reduces the discrepancies between



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(FDA) recommends testing the usability, functionality and satisfaction of new electronic devices as part of the proof of concept.¹⁴ To our knowledge, no technology for direct use by patients has previously been assessed to support MR.

The iatrogenic risk is also increasing with the ageing population and the consumption of multiple, sometimes inappropriate, medications.^{15–17} More studies on experiences in containing the iatrogenic risk are needed. Max is a digital conversational assistant which seems to be straightforward and has been adapted to patients for anaesthesia purposes.¹⁸ Max has also been developed for MR in the clinical pharmacy in our hospital.

Hypothesis

We hypothesise that Max could be used for the first step of MR subject to three factors: (1) that it is well received by patients and has a good compliance rate; (2) that time could be saved in OMS when assisted with Max; and (3) that Max could provide an accurate list of treatments taken at home.

Objectives

The objectives of this observational study were to evaluate in three successive steps the willingness of patients to use Max, the time saved for the OMS when using Max, and the reliability of Max. Finally, patients were asked for their satisfaction with its usability.

METHODS

We present our observational results from this feasibility study on the use of Max for MR to obtain the list of medications taken by patients at home. Sample size has not been calculated for each step. We decided that the duration of each phase would be 2 months due to the inclusion rate and because of human resource constraints. We could not combine all three phases within the same population as it was too time-consuming for the pharmacists.

Medication reconciliation (MR)

MR was conducted by analysing the differences between the medications taken at home and the FPH, according to guidelines.¹⁹ In order to detect avoidable prescription errors, the MR generated the OMS of all prescribed medications, over the counter (OTC) medications and compliance or adherence to treatment. A minimum of three concordant sources of information were compared and compiled to obtain the OMS.²⁰ The patient is one of the sources. When the OMS is provided, the second step of comparison with the FPH can be performed. If discrepancies are noted, they are analysed. If they are intentional, nothing is done; but if they are unintentional, a pharmaceutical intervention is communicated to the prescriber in order to correct errors or mistakes. This study concerned only the first step of the MR, the OMS. Max was tested to obtain the list of medications actually taken by patients at home compared with a conventional telephone call. Only qualified pharmacists performed and validated the final OMS. The telephone calls were done by a student under the responsibility of the pharmacist or by the pharmacists themselves.

Comorbidity assessments

The Charlson Comorbidity Index ranges from 0 to 37 depending on the weight of comorbidities and is calculated using the computerised care record for the medical history of each patient. Comorbidities provide a score associated with the

prognosis.²¹ This index was calculated to explore comorbidities in our populations.

Max robot

The digital robot Max is an inclusive e-Pathway platform. A chatbot (conversational robot on line) is used to provide patients with a questionnaire which retrieves information about their medication treatments at home. Max Pharmacy chatbot combines various functionalities that facilitate the provision of information, in particular a smart search bar and a drug scanner linked to the national public drug database to facilitate data entry and minimise transcription errors. Max is compliant with the General Data Protection Regulation (GDPR). The data collected in Max are stored on a Health Data Hosting (HDH) certified server.

Patient pathway

All patients with a scheduled hospitalisation in Toulouse University Hospital for maxillofacial, otorhinolaryngology surgery or neuroinflammatory care and who were aged >15 years were included. Fifteen days before the consultation with the anesthesiologist or the date of hospitalisation, the pharmacist contacted the patient (or the parent when the patient was aged <18 years) by telephone. The patients were offered the use of the Max robot. If they accepted, an email was then sent with an active link to Max online. After clicking on the link, the router opens the conversational robot and, after a short welcome and a presentation of the system, the questions patients were asked concerned: housing and living conditions, medical history, allergies, and the name, form, route, interval and dosage of each treatment. As the patient typed the name of the medication, a suggestion from the national medication database appeared on the screen which the patient could then validate. Patients also had the option to scan the medication bar code at home or to take a picture of the prescription sheet and download the file. The majority of questions had a 'yes' or 'no' answer (history, allergy, medical implant, OTC medication) and the minority had a free answer such as the name of the allergic agent, the name of the physician or the name of the current pharmacy. If patients did not want to use Max, the pharmacists conducted MR through a conventional telephone call to gather all the medications taken by the patient and the necessary information mentioned above.

First step: willingness and compliance rate of Max

Data were collected over a 3-month period from 22 December 2020 to 29 March 2021. First, in order to measure the willingness to use Max, patients with a scheduled hospitalisation were asked by telephone whether they would like to use Max. If the response was 'yes', we numbered the volunteers who had responded completely to the chatbot, considering here a compliance rate. If the response was 'no', we asked for the reasons by telephone which were categorised as: no mailing box, no internet connection, no desire or no interest, too complicated so preference for the conventional telephone call and no reason given.

Second step: time evaluation

From 14 April 2021 to 17 June 2021, the time spent during the OMS was investigated. All the purposes of the telephone call needed to complete the OMS with and without Max were timed (propose Max, reminders to complete, details if data were not clear, pharmacy dispensary, physician, and primary care nurse).

Third step: accuracy and reliability of Max

From 14 March 2022 to 18 May 2022, data were collected to explore the data accuracy between the two options (Max source and a conventional telephone call without Max). First, the differences were numbered for each patient between the patient source list (with Max or with a conventional telephone call) and the OMS list from other sources. Differences could be: the name, dose, dosage form, lack of a drug or an extra drug in the list. In order to exclude bias, the study operator for the patient source (first telephone call to propose Max or the conventional telephone call) was different from the operator for the OMS obtained with other sources. Finally, the number of differences was weighted by the number of prescription lines. During the last month of the third step, Max underwent a maintenance downtime which explained the higher number of patients with a conventional telephone call.

Patient satisfaction with usability

Patients who responded positively and tried or succeeded to complete Max were asked to answer a satisfaction questionnaire about its usability in hospital. During the three steps of the study, 121 patients agreed to try to use Max (56, 46 and 19, respectively, in steps 1, 2 and 3). Just before leaving the hospital, the questionnaire was left in the room for them to complete with no bias or influence. Pharmacists came back into the room once completed. Possible answers were: not satisfied, moderately satisfied, neutral, satisfied and very satisfied. The satisfaction questionnaire contained four items on functionality (internet and mail connection, digital device, ease of use and legibility) and five items concerned their trust in the device (relevance of questions, number of questions, trust concerning personal data, time spent to complete and improved care through the device). The maximum satisfaction score was 36 points.

Descriptive analysis

The size of the study was determined by the duration of the observation period of 2–3 months for each stage. We collected data through routine observations. For descriptive analysis, mean (SD) values were recorded, except in the graph where SE of the mean (SEM) is shown.

Quality assurance of steps

Data cleaning was performed with a double control by a second pharmacist at the end of the collection, reading the patients' medical records and MR documents. For the third step, the pharmacist who called the patient was different from the pharmacist who called the other sources to make the OMS, to avoid influencing the measurements.

RESULTS

First step

Willingness to use Max after first telephone call

From 22 December 2020 to 29 March 2021, pharmacists tried to contact 79 patients by telephone to prepare for upcoming hospitalisation. More than 70% of the patients (56/79) agreed to use Max, 1.3% (1/79) could not be reached and 27.8% (22/79) declined the use of Max. The characteristics of the population who responded positively to the use of Max and those who responded negatively are shown in table 1. Those who responded negatively to the use of Max were older with a higher Charlson Comorbidity Index and more prescription lines.

The reasons why patients responded negatively were mainly due to having no mailing box or no access to the internet at

Table 1 Characteristics of the populations who responded positively and negatively to the use of Max

	Positive response to Max	Negative response to Max
No of patients	56	22
Age, mean (SD)	45.5 (17.7)	57.5 (21.6)
Female, n (%)	32 (57.1)	10 (45.5)
Charlson score, mean (SD)	1.3 (1.7) n=48	2.3 (2.5) n=20
No of prescription lines taken by patient, mean (SD)	2.6 (2.8) n=49	3.7 (4.9) n=19
n means that the data were available only for this number of patients.		

home (14/22, 63.6%) and no intention or no desire to use an electronic device (6/22, 27.3%). Two patients (9.1%) provided no reason.

Compliance rate of Max

The majority of patients who agreed to use Max (41/56, 73.2%) completed the entire Max process. Overall, of the population contacted by telephone, 41/79 (51.9%) were able to complete Max fully. The patient flow chart is shown in online supplemental figure 1. With regard to the 15 patients who did not respond entirely to Max, after a reminder telephone call seven (50%) changed their mind and the pharmaceutical interview was conducted by a conventional telephone call. Unfortunately, six of the 15 patients (40%) could not be reached by telephone for a reminder and did not respond spontaneously to Max, so they were withdrawn from the study. Two of the 15 patients (13%) failed to complete the process due to technical and connection problems.

Second step: time saved using Max

For two additional successive months, 60 patients were enrolled to explore the time spent for OMS (OMS is the result of three concordant sources of information) with and without Max. Six patients were lost to follow-up before the end of the step. The flowchart shown in online supplemental figure 2 explains the reasons for the withdrawal of patients. The mean time spent for Max-assisted OMS was 13.2 min (SEM 1.1) compared with 20.8 min (SEM 2.8) without Max (figure 1), so Max saved 7 min in our feasibility study. The characteristics of the population who participated in the second step of the study are shown in table 2.

As for the first step, we noted that age, the Charlson Comorbidity Index and the number of prescription lines was higher in the population with a negative response to Max.

Third step: Max accuracy or a conventional phone call for the patient source

For two additional successive months, 46 patients were enrolled in the third step, of whom 19 used Max for the list of drugs and 27 used a conventional telephone call for the patient list of drugs (a maintenance downtime explains the higher number of patients with a conventional telephone call). A flowchart is shown in online supplemental figure 3. The accuracy was examined based on the number of differences between the list of drugs from the patient source (with Max or conventional telephone call) and the list obtained from other sources. The number of differences (3.4 vs 3.0) and the number of differences divided by the number of prescription lines (0.7 vs 0.5, respectively, in the Max group and the conventional telephone call group) were higher in the

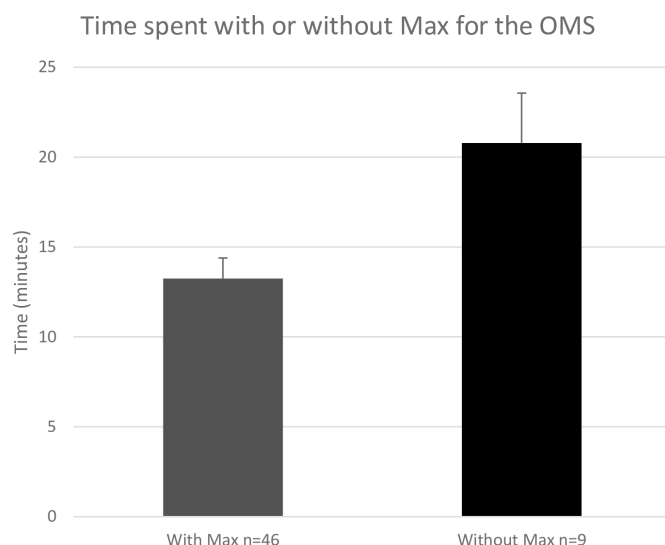


Figure 1 Time saved for the optimised medical synthesis (OMS) with Max. Data shown are mean and SE of the mean (SEM).

Max group. The population characteristics according to Max or conventional telephone call are shown in table 3.

Patient satisfaction with usability

Satisfaction questionnaires were supposed to be filled out before discharge, independently, when patients agreed to use Max, if they completed Max entirely, or even if they tried but failed to complete entirely. Fifty-four of 121 (44.6%) responses have been analysed. The proportion of patients satisfied with Max is shown in figure 2. Only five patients were dissatisfied with the number of questions and three with the time spent to complete Max. All the patients were satisfied with the ease of use. The mean score was 30.9/36 (SD 4.7).

DISCUSSION

To the best of our knowledge, the patient source support tool Max has never been tested for MR. Globally, Max was well received during the first contact with patients by telephone. 70% of the patients were willing to use Max to prepare their hospitalisation during a pharmacy MR process. In the literature, Max adapted for use in cataract surgery or anaesthesia (from the same publisher Bot design) seems also to be well accepted by patients.^{18 22} For middle-aged patients, these results conform with the literature concerning the connected tool Medtrue for preparing MR in primary care, for example.²³ Older adults

Table 2 Characteristics of the populations who gave positive or negative responses to testing Max

	Positive response to Max	Negative response to Max
No of patients	46	9
Age, mean (SD)	42.8 (17.9)	58.1 (15.9)
Female, n (%)	25 (54.3)	6 (66.7)
Charlson score, mean (SD)	1.4 (1.7) n=28	2.7 (2.0) n=6
No of prescription lines taken by patient, mean (SD)	3.2 (3.2) n=34	4.3 (2.6) n=8
Time spent for OMS, min, mean (SD)	13.2 (7.8) n=45	20.8 (8.3) n=9
OMS, optimised medical synthesis.		

Table 3 Characteristics of the two populations according to Max use or conventional telephone call

	Max for patient's drug list	Conventional telephone call for patient's drug list
No of patients	19	27
Age, mean (SD)	49.7 (15.8)	59.2 (16.0)
Female, n (%)	8 (42.1)	11 (40.7)
Charlson comorbidity score, mean (SD)	2.2 (1.5)	3.2 (2.0) n=22
No of prescription lines taken by patient, mean (SD)	4.5 (2.8)	6.2 (3.9)
No of differences between the patient list and the OMS list of drugs, mean (SD)	3.4 (3.4)	3.0 (2.9)
No of differences divided by number of prescription lines, mean (SD)	0.7 (0.4)	0.5 (0.4)
OMS, optimised medical synthesis.		

should benefit from MR more than other patients because the link between age and discrepancies has been shown by simple and multivariate analyses.^{24 25} Technical and human obstacles tend to decrease with new generations of older adults who have usually experienced connected devices during their early ageing. Moreover, adapted wireless tools are created to improve connections in this population.²⁶ We cannot draw conclusions from our study regarding this specifically aged population and the use of Max as it mostly involved young volunteers.

Pharmacy-led MR seems to be an effective strategy in meta-analysis.⁵ For the first time, we examined Max as a patient chatbot that could save 7 min in MR but randomised controlled studies are needed to confirm this trend. The assumption is that MR can take 20–79 min depending on the participants, patients, diseases and the availability of primary care professionals.⁹ Depending on the reference, 7 min can represent 8–33% of MR time. It seems low, but human resources are always limited and connected tools will probably be helpful. Moreover, the quality of the medication list from Max must be weighed against the time saved.

In the emergency setting, 70% of care providers state that during MR the information from patients during a face-to-face interview is the most relevant and precise.²⁷ Collecting information with an electronic tool such as Max rather than a human professional may involve risks related to the quality

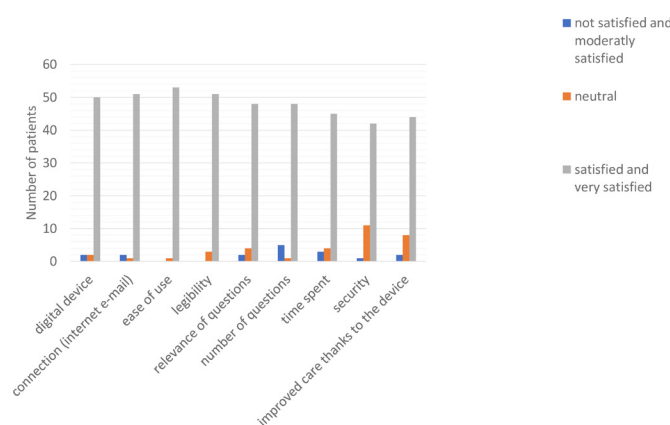


Figure 2 Patient satisfaction with usability of Max. The numbers of patients are shown according to the questions. The three categories are not satisfied and moderately satisfied; neutral; and satisfied and very satisfied.

of information collected. Indeed, when patients are invited to update their medication data in a personal digitalised health record, some discrepancies were found with a pharmacy-led MR.²⁸ It remains necessary to assess the lack of human interaction in practice and the issues of responsibility in cases where incorrect information is provided by the patient. Our initial reliability showed a trend towards more differences with Max and, in view of this trend and the inclusion methods that make our populations poorly comparable, self-reported information about medication should be double-controlled on patient admission to hospital.

Limitations

Using data that were not created to answer the specific research question, without randomization and in only one centre, could create bias. Well-conducted studies are needed.

CONCLUSION

This study with Max allows us to consider the use of a digital assistant for carrying out MR. This tool appears to be usable as it is well accepted by patients and could save time, a benefit that remains to be confirmed through a randomised study. However, the tool seems less reliable in terms of patient-sourced information compared with conventional telephone calls.

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